ISOTRETINOIN

Highly teratogenic — Pregnancy Prevention Programme

Mood changes

CHEMICAL NAME (2Z,4E,6E,8E)-3,7-dimethyl-9-(2,6,6-trimethylcyclohex-1-en-1-yl)nona-2,4,6,8-tetraenoic acid (13-cis-retinoic acid)

CLASS Oral retinoid (severe acne)

TRADE NAMES Roaccutane · Accutane · Absorica · Isotroin

THERAPEUTIC USES / INDICATIONS

- Severe nodulocystic acne or acne resistant to adequate courses of systemic antibiotics and topical therapy.
- Acne causing scarring or significant psychological distress (specialist).

ADMINISTRATION

- **PO** capsules with food (fatty meal ↑ absorption), once or twice daily.
- Prescribed by specialist (dermatologist).
- Females: Pregnancy Prevention Programme — pregnancy tests before, monthly and 1 month after; effective contraception 1 month before, during and 1 month after.
- Do not donate blood during and for 1 month after.

DOSAGES

- **Start:** 0.5 mg/kg/day, adjust to 0.5–1 mg/kg/day.
- **Cumulative target:** 120–150 mg/kg per course (usually 16–24 weeks).
- Low-dose regimens (e.g. 0.25 mg/kg/day) for tolerance.

PHARMACOKINETICS

- **Peak:** 2–4 h; bioavailability doubled with food.
- Metabolised in liver (CYP2C8, 3A4) to active 4-oxo-isotretinoin.
- Excreted in urine and faeces. **t½:** 10–20 h.

PHARMACODYNAMICS

- Shrinks **sebaceous glands** and ↓ sebum production by up to 90%; normalises follicular keratinisation; reduces *C. acnes* and inflammation.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dry lips (cheilitis), dry skin, dry eyes, nosebleeds.
- Muscle and joint aches.
- Photosensitivity, hair thinning, initial acne flare.

ADVERSE EFFECTS (SERIOUS)

- Teratogenicity (craniofacial, cardiac, CNS malformations; miscarriage).
- Depression, mood changes, suicidal ideation, psychosis.
- Raised triglycerides/cholesterol, hepatitis; pancreatitis.
- Benign intracranial hypertension (with tetracyclines); night blindness; IBD (debated).

CONTRAINDICATIONS & CAUTIONS

- Pregnancy and breastfeeding; women not adhering to PPP.
- Hepatic impairment; hyperlipidaemia; hypervitaminosis A.
- Use with tetracyclines.

KEY INTERACTIONS

- Tetracyclines (doxycycline, minocycline): intracranial hypertension; contraindicated.
- **Vitamin A supplements:** toxicity.
- **Progesterone-only minipill:** may be ineffective — use more reliable method.

NURSING MANAGEMENT

- PPP documentation, pregnancy tests (females).
- Baseline and 1-month LFTs and fasting lipids, then 3-monthly.
- Ask about mood at every visit (patient and family).
- Lip balm, emollients, eye lubricants advice.

PATIENT EDUCATION

- Never get pregnant during treatment or for 1 month after; use effective contraception.
- Report low mood, irritability, thoughts of self-harm, severe headache or visual changes immediately.
- Use lip balm and moisturiser; avoid waxing, laser, dermabrasion for 6 months.
- Do not share capsules or donate blood.

KEY POINTS

- Oral retinoid; shrinks sebaceous glands.
- Highly teratogenic — PPP.
- 0.5–1 mg/kg/day, cumulative 120–150 mg/kg.
- Mood changes; dryness; lipids/LFTs; no tetracyclines.